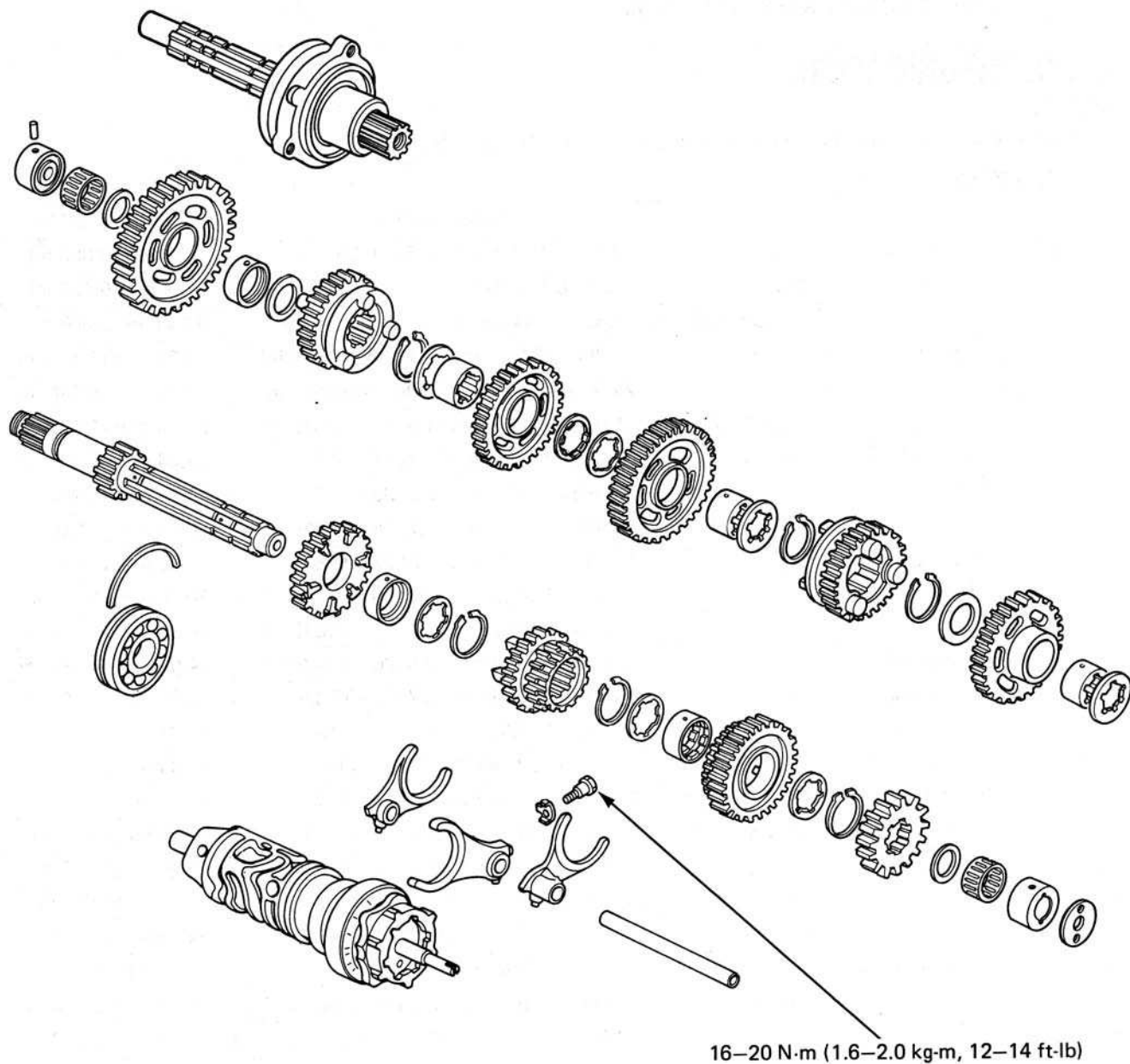




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11 TRANSMISSION





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SERVICE INFORMATION

GENERAL

For internal transmission repairs, the crankcase must be separated (Section 10)

SPECIFICATIONS

			STANDARD	SERVICE LIMITS
Transmission	Backlash	1st, 2nd	0.095-0.190 mm (0.0037-0.0075 in)	0.20 mm (0.008 in)
		3rd	0.094-0.188 mm (0.0037-0.0074 in)	0.19 mm (0.007 in)
		4th, 5th, 6th	0.068-0.136 mm (0.0027-0.0054 in)	0.14 mm (0.006 in)
	Gear I.D.	M5, M6	28.000-28.021 mm (1.1024-1.1032 in)	28.04 mm (1.104 in)
		C1	24.000-24.021 mm (0.9449-0.9457 in)	24.04 mm (0.946 in)
		C2, C3, C4	31.000-31.025 mm (1.2205-1.2215 in)	31.04mm (1.222 in)
	Gear Bushing O.D.	M5, M6	27.959-27.980 mm (1.1007-1.1016 in)	27.94 mm (1.100 in)
		C1	23.959-23.980 mm (0.9433-0.9441 in)	23.94 mm (0.943 in)
		C2, C3, C4	30.950-30.975 mm (1.2185-1.2195 in)	30.93 mm (1.218 in)
	Gear Bushing I.D.	M5	24.985-25.006 mm (0.9837-0.9845 in)	25.03 mm (0.985 in)
		C1	20.016-20.037 mm (0.7880-0.7889 in)	20.06 mm (0.790 in)
		C2	27.985-28.006 mm (1.1018-1.1026 in)	28.03 mm (1.104 in)
	Mainshaft O.D.	at M5	24.959-24.980 mm (0.9826-0.9835 in)	24.94 mm (0.982 in)
	Countershaft O.D.	at C1	19.987-20.000 mm (0.7869-0.7874 in)	19.97 mm (0.786 in)
		at C2	27.967-27.980 mm (1.1011-1.1016 in)	27.94 mm (1.100 in)
Shift Fork	Gear-to-bushing clearance	M5, M6	0.020-0.062 mm (0.0008-0.0024 in)	0.10 mm (0.004 in)
		C1	0.020-0.062 mm (0.0008-0.0024 in)	0.10 mm (0.004 in)
		C2, C3, C4	0.025-0.075 mm (0.0010-0.0030 in)	0.11 mm (0.004 in)
	Bushing-to-shaft clearance	M5	0.005-0.047 mm (0.002-0.0019 in)	0.08 mm (0.003 in)
		C1	0.016-0.050 mm (0.0006-0.0020 in)	0.09 mm (0.004 in)
		C2	0.005-0.039 mm (0.0002-0.0015 in)	0.08 mm (0.003 in)
	Claw thickness		6.43-6.50 mm (0.253-0.256 in)	6.1 mm (0.24 in)
	I.D.	Left and right	14.000-14.021 mm (0.5512-0.5505 in)	14.04 mm (0.553 in)
Fork shaft	O.D.		13.966-13.984 mm (0.5498-0.5505 in)	13.90 mm (0.547 in)

TORQUE VALUE

Centre shift fork 16-20 Nm (1.6-2.0 kg.m, 12-14 ft.lb)

TOOLS

Common

Driver 07746-0030100
Attachment, I.D. 25 mm 07746-0030200



TROUBLESHOOTING

Hard to shift

1. Clutch slave cylinder sticking
2. Shift fork bent
3. Shift shaft bent
4. Shift claw bent
5. Shift drum cam grooves damaged

Transmission jumps out of gear

1. Gear dogs worn
2. Shift shaft bent
3. Shift drum stopper bent
4. Shift forks bent



TRANSMISSION DISASSEMBLY

Separate the crankcase (Section 10)
Inspect the backlash of each gear

SERVICE LIMIT

1st, 2nd	0.20 mm (0.008 in)
3rd	0.19 mm (0.007 in)
4th, 5th, 6th	0.14 mm (0.006 in)

Remove and disassemble the mainshaft and
countershaft

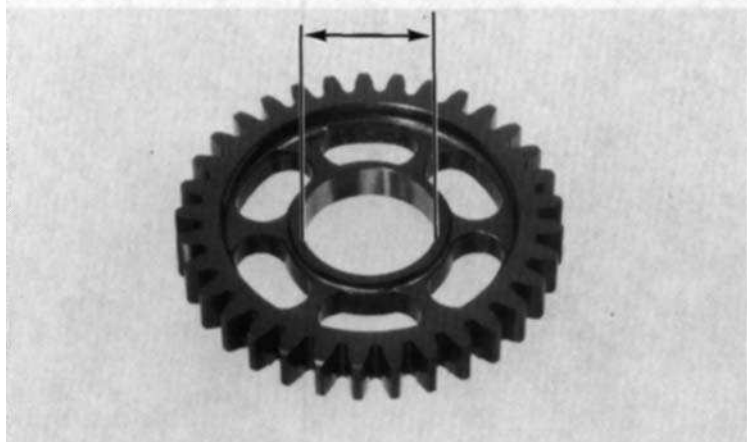
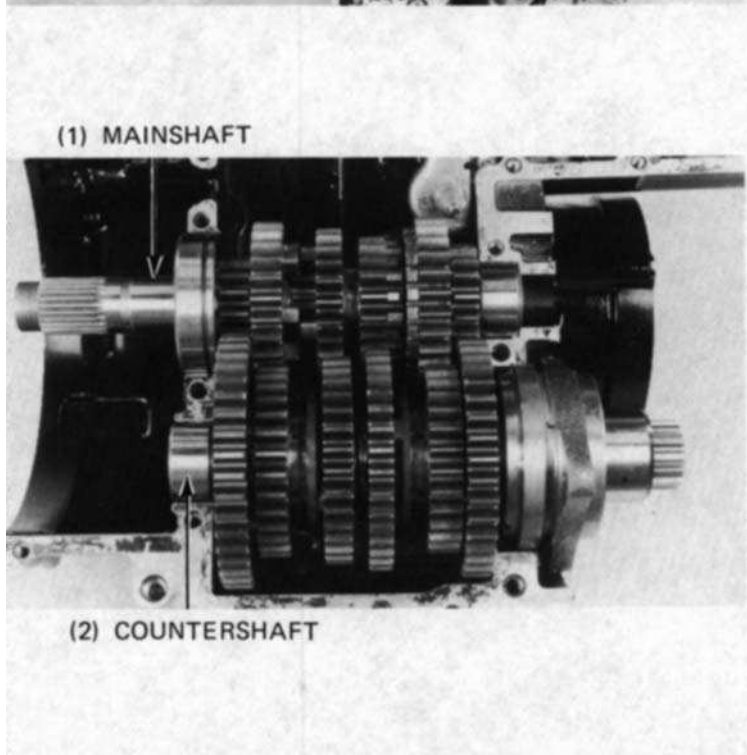
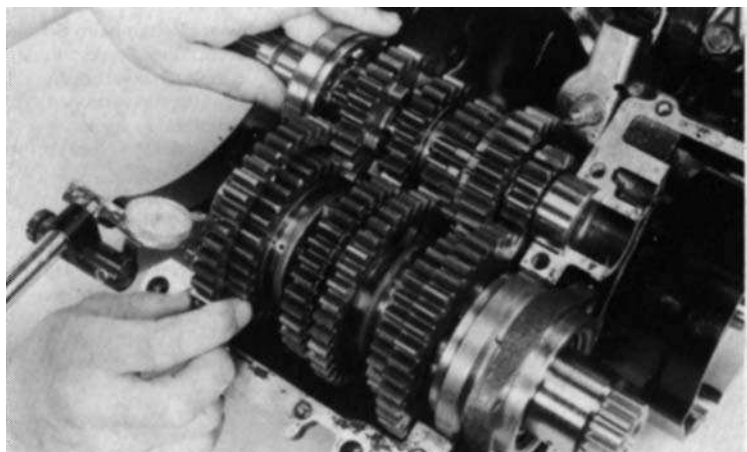
INSPECTION

Check the gear dogs, dog holes and teeth
for excessive or abnormal wear, or evidence
of insufficient lubrication.

Measure the I.D. of each gear.

SERVICE LIMITS:

M5, M6	28.04 mm (1.104 in)
C1	24.04 mm (0.946 in)
C2, C3, C4	31.04mm (1.222 in)





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Measure the O.D. of each gear bushing

SERVICE LIMITS:

M5, M6:	27.94 mm (1.100 in)
C1:	23.94 mm (0.943 in)
C2, C3, C4:	30.93 mm (1.218 in)

Calculate the clearance between the gear and bushing.

SERVICE LIMITS:

M5, M6:	0.10 mm (0.004 in)
C1:	0.10 mm (0.004 in)
C2, C3, C4:	0.11 mm (0.004 in)

Measure the I.D. of gear bushings.

SERVICE LIMITS:

M5:	25.03 mm (0.985 in)
C1:	20.06 mm (0.790 in)
C2:	28.03 mm (1.104 in)

Measure the O.D. of the mainshaft and countershaft.

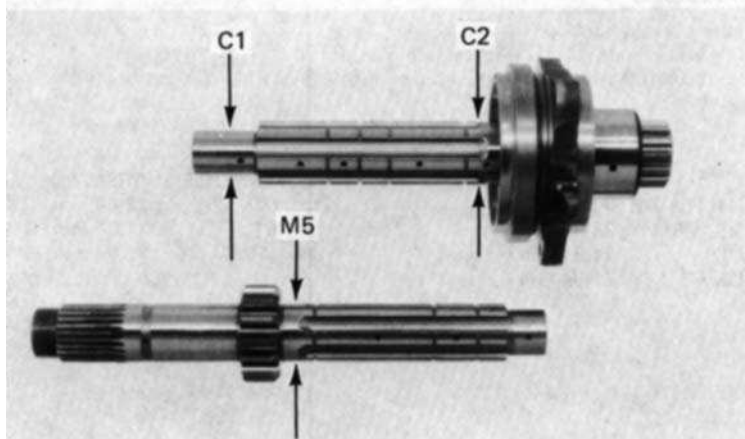
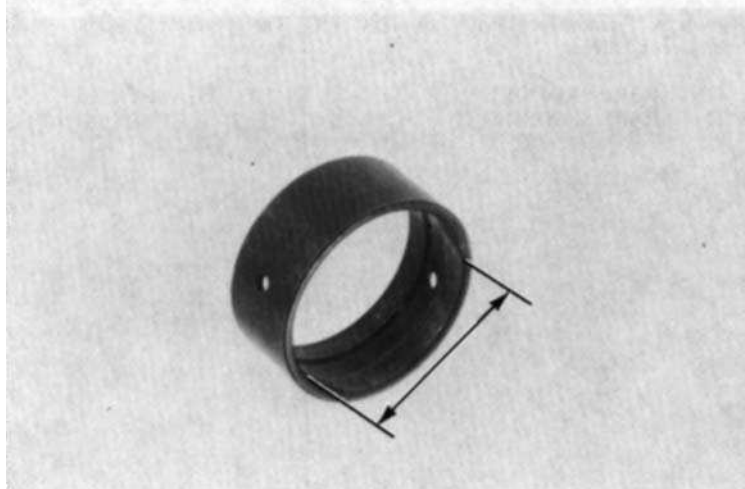
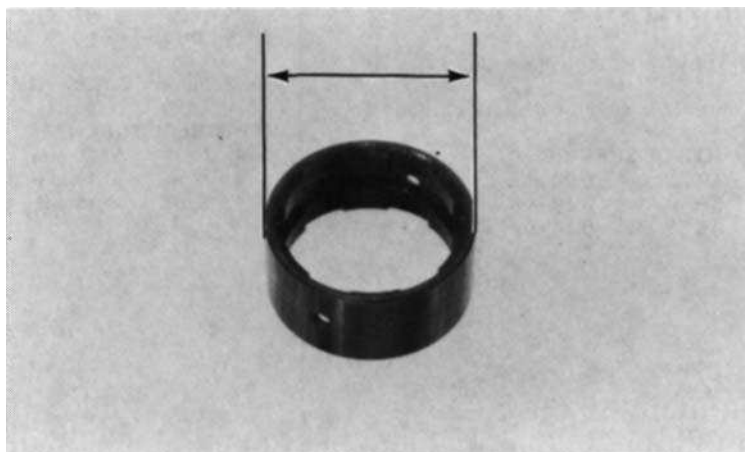
SERVICE LIMITS:

MAINSHAFT (at M5 bushing)	24.94 mm (0.982 in)
COUNTERSHAFT⊗at C1 bushing)	19.97 mm (0.786 in)
COUNTERSHAFT (at C2 bushing)	27.94 mm (1.100 in)

Calculate the clearance between the bushing and the shaft.

SERVICE LIMITS:

M5:	0.08 mm (0.003 in)
C1:	0.09 mm (0.004 in)
C2:	0.08 mm (0.003 in)





TRANSMISSION ASSEMBLY

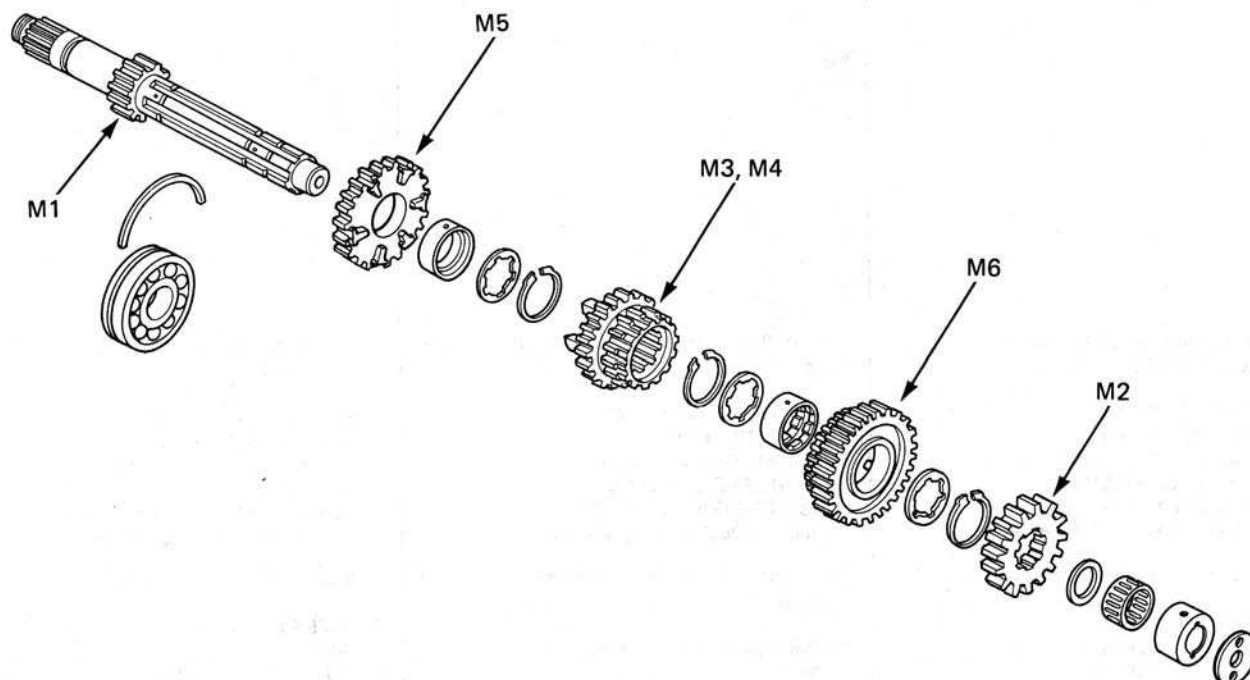
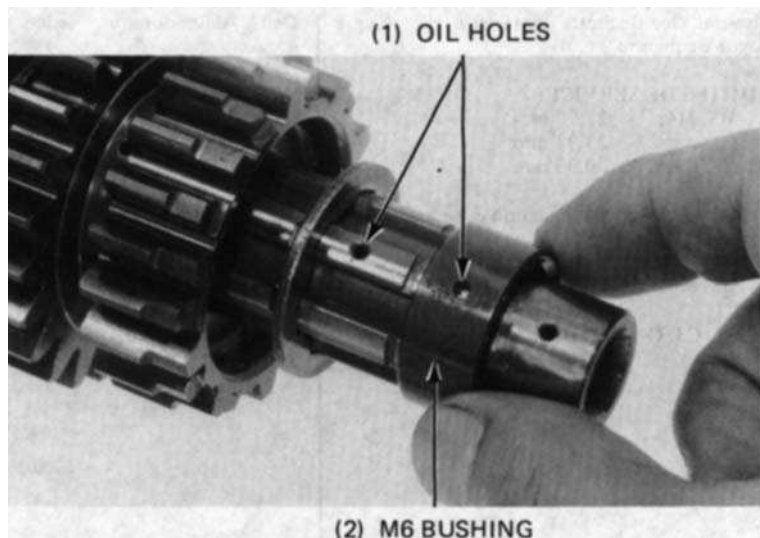
MAINSHAFT

When the mainshaft ball bearing was removed, drive a new bearing using driver (07746-0030100) and I.D. 25 mm attachment (07746-0030200).

Check the gears for freedom of movement or rotation on the shaft.
Check that the snap rings are seated on the grooves and align their end gaps with the lands on the splines.

NOTE

Align the oil holes in the M6 bushing and the shaft.





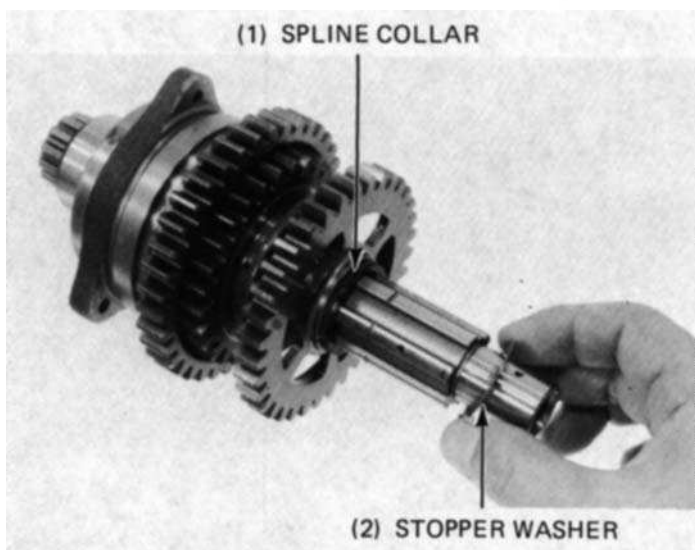
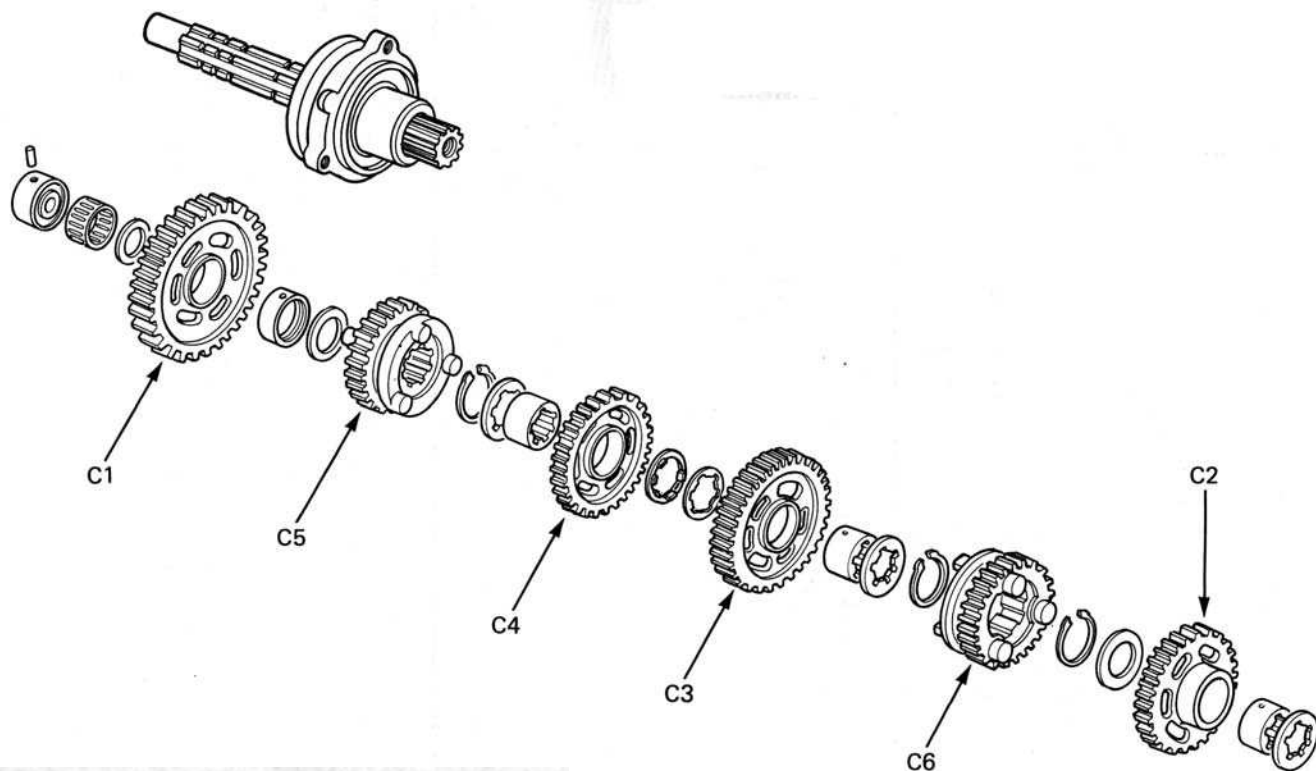
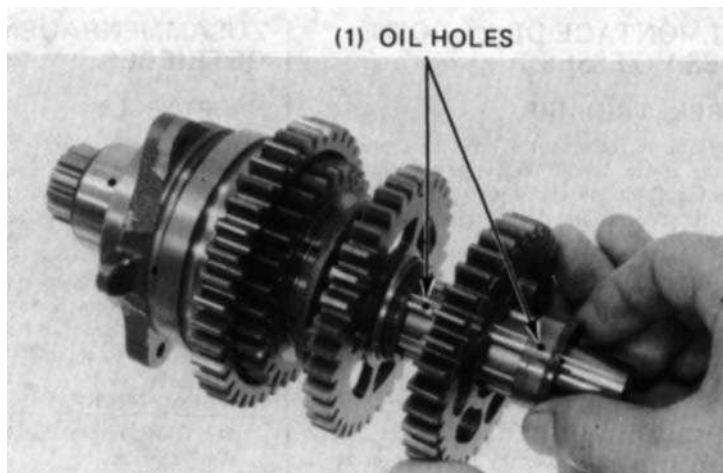
COUNTERSHAFT

Check the gear for freedom of movement or rotation on the shaft.

Check that the snap rings are seated in the grooves and align their ends with the lands of the splines.

NOTE

Align the oil holes in the C3 and C4 bushings, and the shaft.

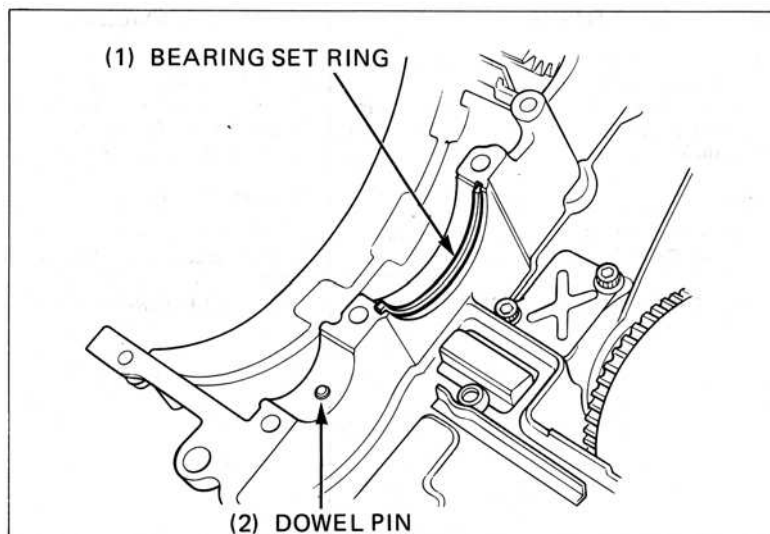


NOTE:

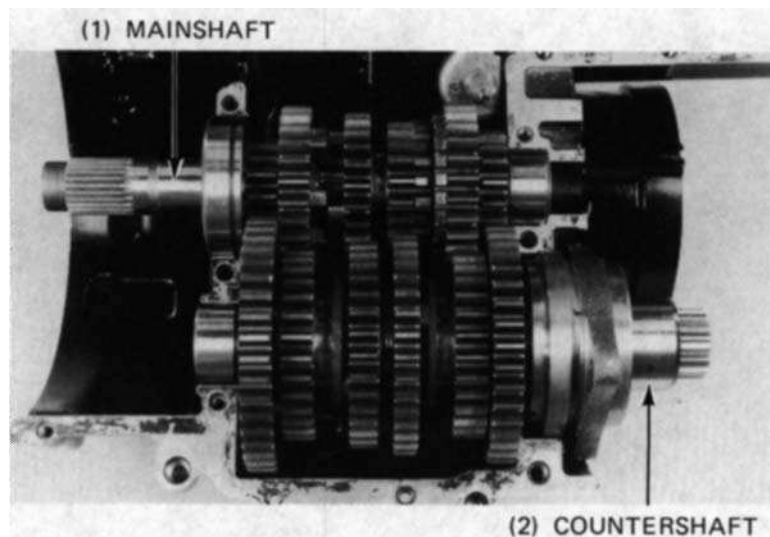
Install the stopper washer while aligning the tabs of the washer with the grooves in the spline collar.



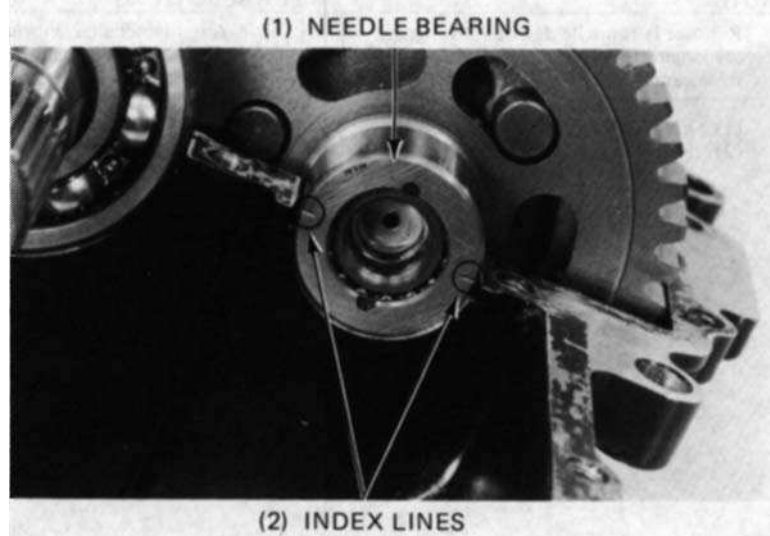
Install the mainshaft bearing set ring and countershaft bearing dowel pin into the upper crankcase.



Install the mainshaft, aligning the bearing grooves with the bearing set ring. Install the countershaft.



Align the index lines on the countershaft needle bearing with the crankcase mating surfaces to insert the dowel pin into the needle bearing hole.

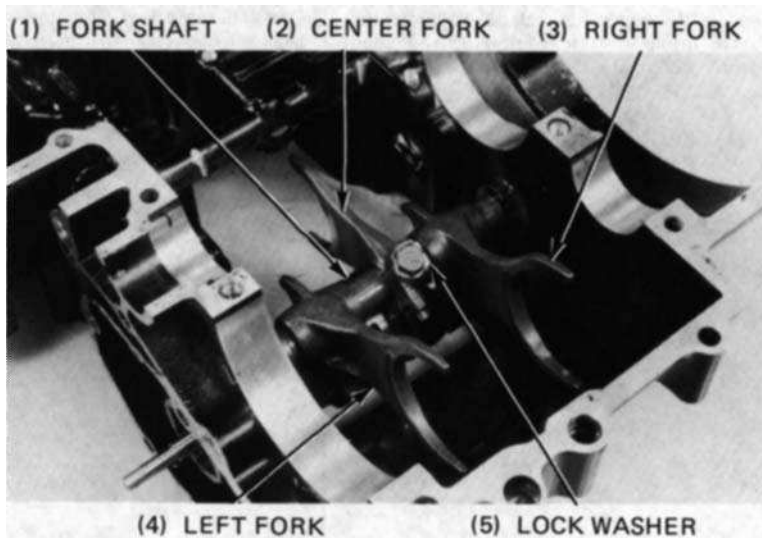




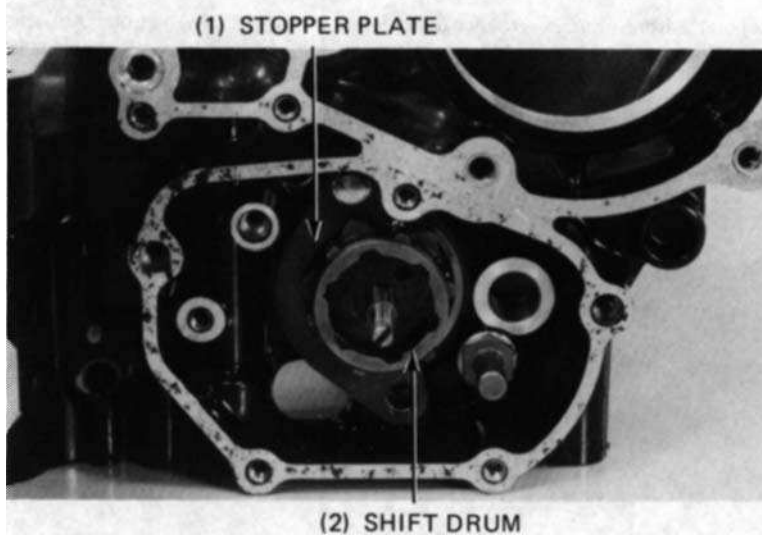
SHIFT FORK AND SHIFT DRUM

REMOVAL

Bend the lock washer tab down and remove the centre fork mounting bolt. Remove the shift fork shafts and shift forks.

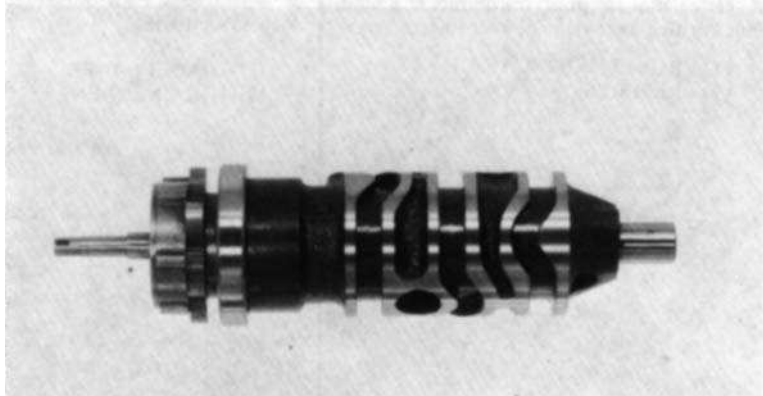


Remove the shift drum bearing stopper plate.
Remove the shift drum.



GEAR SHIFT DRUM AND SHIFT FORK INSPECTION

Inspect the shift drum end for scoring, scratches, or evidence of insufficient lubrication.
Check the shift drum grooves for damage.





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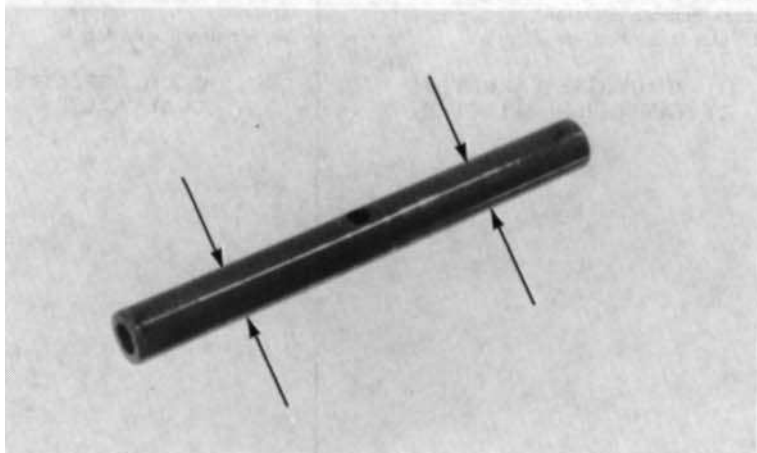
Inspect the shift drum and shift fork shaft holes for scoring or scratches.



(2) SHIFT DRUM HOLE

Measure the shift fork shaft O.D. where the shift forks move on the shaft.
Check for scratches, scoring or evidence of insufficient lubrication.

SERVICE LIMIT: 13.90 mm (0.547 in)

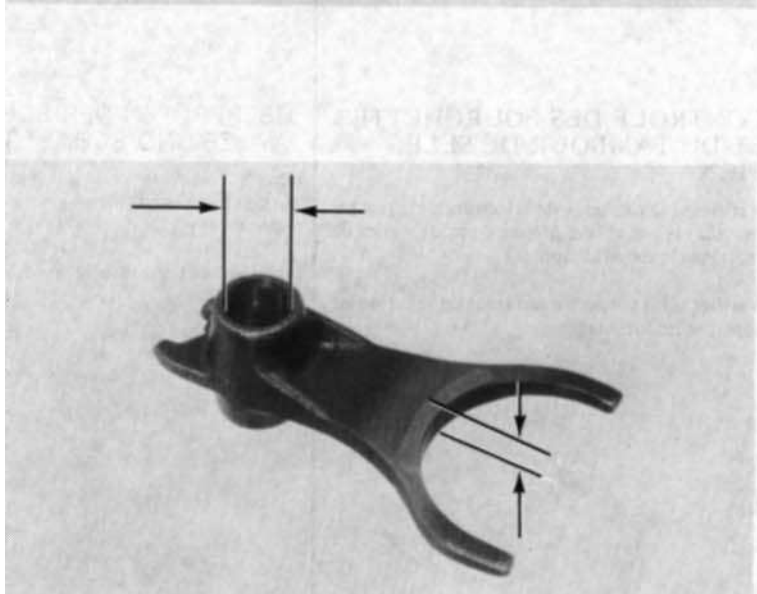


Measure the right and left shift fork I.D.
Measure the shift fork claw thickness.

SERVICE LIMITS:

I.D. (left and right fork): 14.04 mm (0.553 in)

CLAW THICKNESS: 6.1 mm (0.24 in)



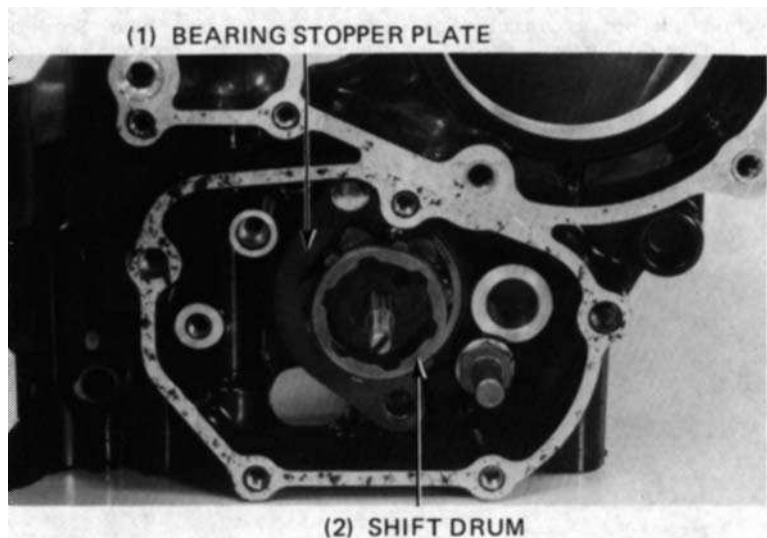


INSTALLATION.

Install the shift drum.

Install the dowel pin into the lower bolt hole of the bearing stopper plate.

Apply locking agent to the stopper plate bolt threads and install the stopper plate.
Remove the dowel pin.



Install the shift fork shaft and shift forks.
Install a new lock washer and the bolt to the centre shift fork and tighten the bolt.

TORQUE:

16-20 Nm (1.6-2.0 kg.m, 12-14 ft.lb)

Bend up the lock washer's tab.

